

Agricultural ICT Case Studies

Computing and Mathematics

May, 2026

Agricultural ICT Case Studies (A21681)

Short Title: Agricultural ICT Case Studies
Faculty Science and Computing
Department Computing and Mathematics
Cluster AgriFoodICT
Author UMCCARTHY
Credits 5

Level: Advanced

Description of Module / Aims

This module aims to provide students with an understanding of state of the art tools and technologies at currently being deployed at work in the industry. This course will involve a number of relevant guest speakers discussing both experience and presenting case studies as well as a number of field trips to relevant stakeholders.

Programmes

COMP-0979	HDip in Science in Agri-Food ICT Systems (WD_SAFICT_G)	5 / 1 / M
	Higher Diploma in Science in Agri-Food ICT Systems (WD_18AFICT_G)	1 / 1 / M

Indicative Content

- Evolution of ICT in the Agri Food Sector
- Guest Lecturers and Seminars
- State of the art in Agri Food ICT Application
- Sensor Technology and connectivity application and demonstrations

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise current best practice within the Agri Food sector.
2. Compare different capabilities of sensing technologies across the sector.
3. Assess the principles of Agri Food ICT technologies.
4. Demonstrate methodologies and techniques for the analysis of Agri Data to produce reports based on seminars and field trips.
5. Discuss the evolution of ICT in the Agri Food Sector.
6. Explore ICT services and technologies from the perspective of security, privacy and cost.
7. Compare and contrast the main technologies used in the Agri Food sector.

Learning and Teaching Methods

- This module will be presented by a combination of lectures, guest lectures, practicals and field trips.
- The lectures will be used to introduce new topics and their related content.
- Many topics will be supported with case studies and examples.
- The student will be encouraged to participate in class discussions ask questions to support their learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5,6,7
<i>Project</i>	50%	1,2,4,6
<i>Assignment</i>	20%	4,6,7
<i>In-Class Assessment</i>	30%	1,2,3,5,7

Assessment Criteria

<40%: Inadequate achievement of the learning outcomes.

40%--49%: Will be able to show a basic level of achievement in relation to the key learning outcomes.
Relatively poor analytical and communication skills.

50%--59%: Reasonable achievement of learning outcomes. Good analytical and communication skills.

60%--69%: In addition, the student will demonstrate more detailed achievements of project goals
including demonstrating an ability to evaluate knowledge acquired.

70%--100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all
learning outcomes, predict general trends in behaviour including exceptional behaviour, and
critically evaluate and relate topics.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Requested Resources

- Lecture Room: Loose Seated